SKINNER'S SCIENTIFIC SPECTRUM

"*The universe is a deeply unusual thing.*

- *Neo Skinner*

Mathematics

Maths Has Touchdown

In our distinctly modern world, with such glorious things as pocket calculators and portable computers, it is sometimes difficult to remember that mathematics was not always widely known. Ask a medieval peasant to multiply 6 and 7 and the answer they would probably give you is almost certainly not 42. So how did mathematics reach England?

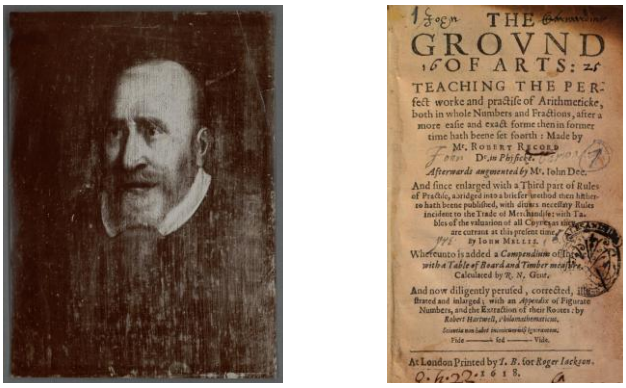
Mathematics is one of the staples of a developed society and we have evidence of pseudo-mathematical thinking as far back as 3000 BCE and even further if you consider the geometric patterns used in the Ubaid culture in 5000 BCE or the use of money-like tokens by Neolithic man in 8000 BCE (Ebrahim, 2023). Logic and reasoning are not by any definition new phenomena and the earliest named mathematician known today came with Ahmose, an ancient Egyptian scribe from roughly 1550 BCE who wrote the Rhind papyrus – an ancient Egyptian mathematical textbook measuring over 50 cm long (University of Waterloo, 2024; British Museum, n.d.)!

Mathematics, Read It, Mate! (Section of the Rhind Papyrus)

Since the ancient Egyptians, many other mathematicians have written down their work on paper providing us with a list of mathematicians running all the way from Ahmose to the main subject of this article: Robert Recorde.

Recorde was the mathematician who finally brought the medieval peasants arithmetic. Well, not quite, books were at that point still a commodity for the nobility – even after Gutenberg invented that device we now curse at on a seemingly daily basis. Robert Recorde published the first English arithmetic book in 1543: *The Ground of Arts* was different to other mathematical books of the time, it was written in a language other than Greek and Latin, enabling an entirely new class of people to learn about arithmetic.

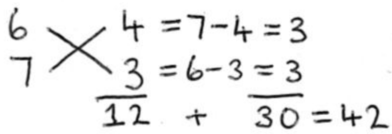
The book is written as a conversation between a master and a scholar and Recorde explains not only how to perform the basic calculations of arithmetic but also how to apply them to the real world featuring problems involving cattle and money – real problems for real everyday people. That was the brilliance of Recorde, he wrote about concepts shared between the nobility and made them accessible to the people like never before – it really is a wonder that Pythagoras gets so much credit!



A Proud Father; LTR: Robert Recorde, *the Ground of Arts* Cover Page

Recorde actually presents mathematical methods that we would not recognise today such as his multiplication technique:

To multiply 6 and 7, Recorde suggests writing the two digits next to their tens complement (number bond of 10) – in this case, 4 and 3 – and drawing a cross between them. Then, the units digit is the product of the two right-hand digits $(4 + 3 = 12)$ and the difference between the numbers on either diagonal is the tens digit $(7 - 4 = 3$ or $6 - 3 = 3 \longmapsto 30)$. By adding together the two numbers $(12 + 30)$ we find that the product of 6 and 7 is 42.



So Simple Even a Fool Can Do It

This amazing method even works for numbers greater than 10 – simply use negative numbers on the right-hand side and the same method yields correct results. I will leave it as a task for the reader to find a proof for why this works.

It is now believed that the ‘cross’ drawn in this method is the origin of our modern multiplication symbol, although Recorde did not use the symbol as we would recognise it in his books. It took until John Napier adopted it in 1618 for the multiplication sign to fall into common use. Fun fact: the symbol was actually originally called the ‘cross of San Andreas.’ Recorde also introduced the equals sign in his later book *The Whetstone of Witte* as “noe .2. thynges, can be moare equalle” (Recorde, 1557) than two parallel lines.

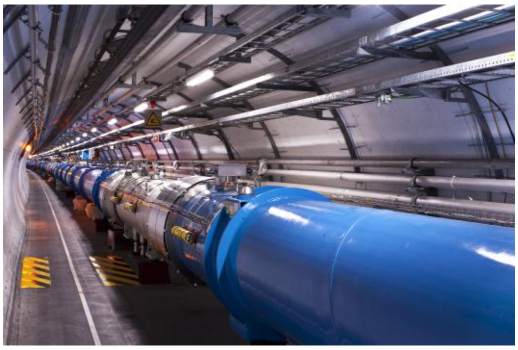
Today, *The Ground of Arts* is a mere curiosity that is only really found in old university libraries. Robert Recorde lacks the recognition that he so rightly deserves in our modern world, replaced by both older mathematicians in Pythagoras and Archimedes, and newer in Leibniz and Euler. But, on our pages, remnants of his great works remain as symbols.

Article By NSR

Physics

Europe Has a Need, a Need for Speed

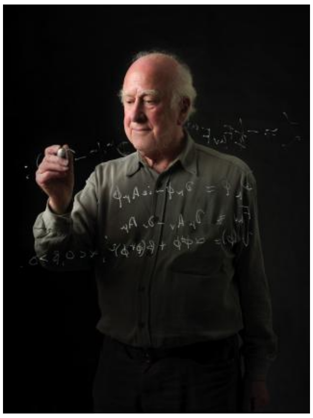
Around and around and around, the world's fastest proton reached 299 780 455 metres per second in the large hadron collider's (LHC) main 27-kilometre-long ring under the idyllic Swiss Alps (CERN, 2008). The journey of this proton lasts for 24 minutes and 20 seconds (CERN, 2024); in this time, it is accelerated to 99.9999991% of the speed of light (CERN, 2008) and comes to a dramatic demise when it smashes head-on into an identical proton travelling in the opposite direction.

Above and Below; LTR: CERN's Base in Meyrin, the Large Hadron Collider

This is one of the many stories of extraordinary human courage and determination from CERN, the European Council for Nuclear Research (Conseil Européen pour la Recherche Nucléaire in original French), who on the 4th of July 2012 discovered the Higgs boson in the large hadron collider – that proton proved very important.

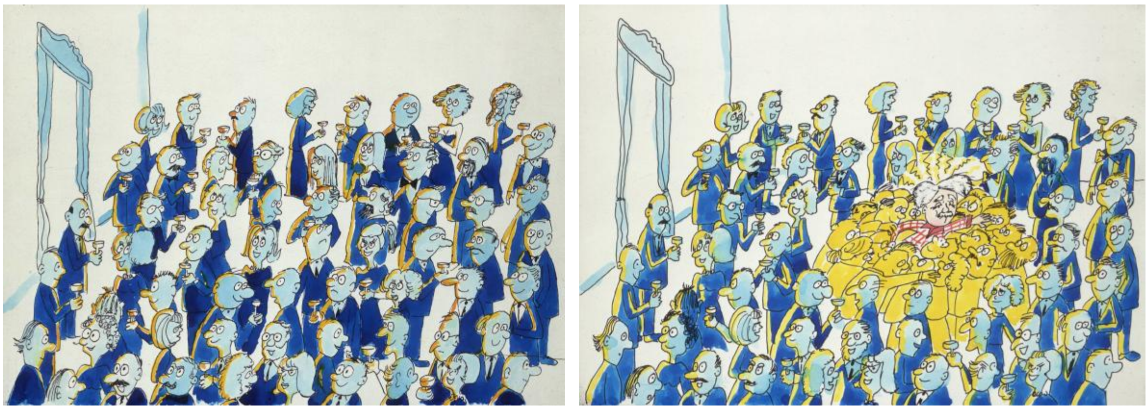
But, hold on, what is the Higgs boson? How is it important? And why was it discovered at CERN?




The Legend of CERN; LTR: Peter Higgs, a Visualisation of the Higgs Boson

Let us rewind back to 1964 when quantum mechanics was still in its infancy, where a problem in physics had just reached the desk of the young physicist Peter Higgs. The issue was with an area known as 'boson symmetry' where the theoretical masses of bosons did not quite line up with those found in nature. And that *quite* might just be the understatement of the century. You see, the problem was that theoretically, bosons had no mass *at all.*

This small issue was solved by multiple physicists at roughly the same time and they said: ‘Invent a new field.’ This field became known as the Brout-Englert-Higgs field (BEH field) after the first three physicists to theorise its existence. The BEH field is central to what is called the Brout-Englert-Higgs mechanism (BEH mechanism) which is a bit of a mouthful to explain, but here goes:



Comic Strip Presents... The BEH Mechanism

Imagine a room full of physicists (illustrated above) – this will be our BEH field – and let us send into the room a hairdresser. Since the hairdresser is unlikely to know any of the physicists or their physics, they pass through the room unnoticed – our hairdresser is a photon with no mass. But, if instead we send in Albert Einstein, he will instantly be surrounded by a cluster of physicists and find it very difficult to move. This difficulty in movement is actually mass in our rather crude model, so this means that particles with more mass interact more with the BEH field and, in fact, it is the field that gives them mass. Bizarre!

Keep that room in your mind for a tiny bit longer, we have one more demonstration to conduct: sending a rumour into the room. Let us imagine that someone enters the room claiming that Galileo Galilei will soon arrive, this causes the physicists to clump around themselves as the rumour travels across the room – this is the Higgs boson, a ripple (excitation) in the BEH field. And it is the Higgs boson that was discovered at CERN.

With that out of the way, how did the fabulous people at CERN prove Higgs right? The large hadron collider is the highest-energy collider anywhere in the world and this means that it can perform experiments in conditions close to those of the very early universe. This is useful, as a high-energy collision was hypothesised to ‘jiggle’ the BEH field to produce an excitation – a Higgs boson. However, there was still one main problem: detecting the new particle. To detect the Higgs boson, physicists at CERN analysed what particles the boson could decay to and recorded which of these were produced in the billions of collisions that occurred in the LHC.

66 years after Peter Higgs first proposed the Higgs boson, it was discovered on the 4th of July 2012 – the biggest event of quantum physics so far this century. That everyone, is the story of the Higgs boson at CERN.

Article By NSR

Chemistry

Cooling from Global Warming?

In mines located in the far-flung region of Inner Mongolia, China continues to mine roughly 1500 Mt (1 megatonne ≘ 1 Tg) of coal each year, accounting for just over 17% of global coal production (IEA, 2023). This coal, once burned, will produce 5.6 Mt of sulfur dioxide – 31.6 Mt globally (Nussey, 2019) – from sulfur impurities that will enter the Earth's atmosphere and then fall as acid rain, lowering the pH of water sources and destroying habitats.

$$\begin{gathered} S_8\ (s) + 8O_2\ (g) \longrightarrow 8SO_2\ (g) \\ 2SO_2\ (g) + O_2\ (g) \longrightarrow 2SO_3\ (g) \\ SO_3\ (g) + H_2O\ (l) \longrightarrow H_2SO_4\ (aq) \end{gathered}$$

Acidifying the Atmosphere, the Production of Acid Rain From Sulfur Impurities

For a brief moment, let us take a look at the effects of sulfur dioxide in the past – 66 million years (Ma) ago to be specific (Keller, 2020). At that point, the Deccan mountain range – a volcanic region located in the Indian subcontinent (still an island for about 16 Ma (Geology Page, 2019)) – was just becoming active and began to spew large amounts of both carbon dioxide and sulfur dioxide. The distribution of the two gases was uneven and led to a period of rapid warming followed by one of equally rapid cooling – how?

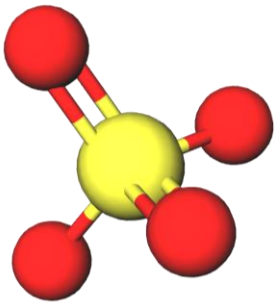
We always imagine that climate change is caused by the major greenhouse gases alone – carbon dioxide and methane – but, many gases actually affect the atmosphere. The carbon dioxide initially caused global warming but then the effects of the sulfur dioxide kicked in with global cooling, you see, before the sulfuric acid falls as rain, it sits as an aqueous mixture of hydronium (H_3O^+) and sulfate (SO_4^{2-}) ions. It is the sulfate ions (or aerosols) that are responsible for the cooling effect, as their bonds act as 'condensation nuclei,' simply: they provide a place and conditions for water vapour to condense. The water then bounces back the sunlight and reduces the amount of energy absorbed by the Earth from the Sun. We label this by saying that sulfate ions have a high albedo, meaning they reflect a large amount of light.

$$\begin{gathered} H_2SO_4 + H_2O \rightleftharpoons H_3O^+ + HSO_4^- \\ HSO_4^- + H_2O \rightleftharpoons H_3O^+ + SO_4^{2-} \\ (H_2SO_4 + 2H_2O \rightleftharpoons 2H_3O^+ + SO_4^{2-}) \end{gathered}$$

Cool – Huh, the Formation of Sulfate Ions

This rapid warming and subsequent cooling was enough to give all life on land a pretty harsh time, let alone the effects of the ash, mercury and hydrochloric acid from the eruption. So much so that it was the supervolcanic eruption that caused this particular mass extinction and *not* the asteroid that we initially assumed wiped clean the face of the Earth.

So, with all that said, why is this climate-focused article set millions of years before the Industrial Revolution? Because the science offers hope. However, the promise of rapid global cooling is still years from being reality and there are some pretty major consequences that humanity would have to face if sulfate aerosols are the answer.



Promises Are Made To Be Broken, the Sulfate Ion (Aerosol)

The first problem is changes in the hydrological cycle, with precipitation more frequent if sulfates are released into the lower atmosphere. This increased precipitation would also be strongly acidic at the concentrations of sulfates required – *yikes.* Some studies even suggest changes in ozone levels and sea-ice levels (Jones, Haywood and Jones, 2016). But, this can be partially avoided by directly injecting sulfates into the upper atmosphere, far above the clouds.

Partially is the key word, though. Injecting sulfates is likely to decrease ozone levels through a complex mixture of aerosol and free radical chemistry. The mechanism runs somewhat like this: sulfate aerosols provide a site for chlorine activation, turning chlorine molecules and ions into chlorine free radicals ($Cl\cdot$). These free radicals then undergo free radical substitution with ozone molecules (O_3) to form oxygen molecules (O_2).

$$\begin{gathered} Cl_2 \xrightarrow{\text{UV}} 2Cl\cdot \\ Cl\cdot + O_3 \longrightarrow ClO\cdot + O_2 \\ ClO\cdot + O_3 \longrightarrow Cl\cdot + 2O_2 \\ (2O_3 \longrightarrow 3O_2) \end{gathered}$$

The Runaway Reaction, Free Radical Substitution

Some scientists (Jones, Haywood and Jones, 2016) have suggested remedies to this ozone depletion conundrum including alternative aerosols such as black carbon (similar to soot) and minerals with high refractive indices. These suggestions work by absorbing sunlight or by the aerosols themselves reflecting light, instead of water droplets. Both of these alternatives would cool the atmosphere but would also cause increased global dimming, leading to something similar to a nuclear winter rather than the rapid cooling clear skies offered by sulfates. However, the other major problem with a geoengineering scheme using aerosols is producing the amount required as it is likely that billions of tonnes of aerosols will be needed and they cannot simply be left to drift into the air, they need to be released at seriously high altitudes!

Whether a method of geoengineering ultimately proves successful in reducing our global temperature to reverse climate change, one thing is certain: we need to reduce carbon emissions. All strategies to reduce or reverse the effects of climate change would need to be continued indefinitely if international agreements to end carbon emissions are not found – and that is truly unsustainable.

Article By NSR

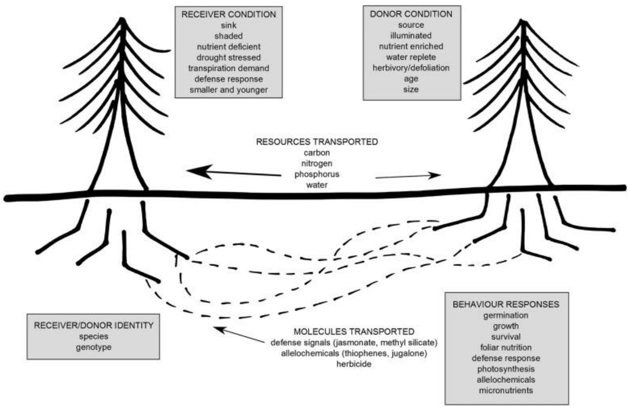
Biology

All Aboard the Fungal Underground!

Picture a misty forest, the leafy tranquillity only disturbed by the flutter of birds in the canopy above. Light dapples the bluebells carpeting the forest floor as the morning dew trickles down stems and seeps into the soil. Beneath the litter, you will find a bustling highway of communication, trade and war between the Davids and Goliaths of the forest. This is the mycorrhiza or mycorrhizal network (MN) found within and between fungi, myco-, and 90% of plants' roots, -rhiza (ScienceDirect, 2023b).

Like all things ecological, this relationship is a symbiosis. Fungal hyphae can access soil space that root hairs cannot and supply mineral nutrients (especially phosphate and nitrate) to their hosts, improving plant health and fitness. Hosts exploit this fungal-optic cabling to transfer nutrients with their neighbours. Yet, the fungi tax as much as 30% of the sugar (Grant, 2018), and nitrogen (as amino acids), transferred between host plants, in exchange for their services.

Generally, flows occur from the fittest to the weakest due to seasonal variations in photosynthetic output, disease, pests etc. One of the most researched advective (large-scale) transfers involves mature 'hub' trees which, being the oldest and wisest, have the largest fungal network. These hub trees use their deep roots and large canopy to transfer energy, nutrients and water to saplings that receive only 3% of the light as the canopy (Grant, 2018). Douglas fir trees are known to recognise each other and prioritise carbon flow accordingly, upping the flow when they detect allelochemical 'distress signals.'

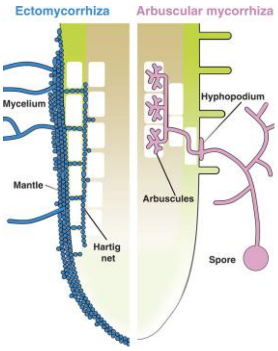


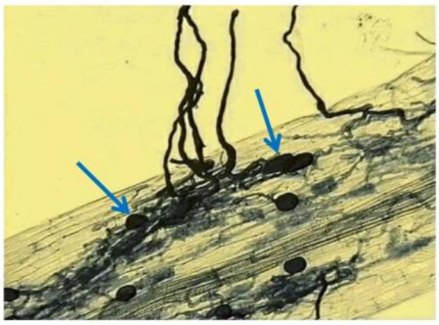
LTR: The Who, What, Where, When, and Why of the MN; a Map of the Jubi-Tree Line

Warnings can also be sent via nature's 'Hy-Fi.' For example, Song et al. (2014) showed that tomato plants infested with leaf-eating caterpillars passed jasmonate (lipid-based phytohormones) SOSs through the MN to their neighbours, who exhibited up-regulation of four defence-related genes only 6 hours after infestation of the 'donor' plants. Plants have even developed chemicals to promote mycorrhizal fungi, one of the most significant being strigolactones – cyclic esters derived from carotenoids. Their existence emphasises the importance of the MN to individual and community-wide health.

One of the most common methods of investigating MN transfers is through tracer studies. For example, forest ecologist Allen Larocque tracked the distinctive 'salmon nitrogen' from fish carcasses left by bears through trees away from the banks of salmon streams shared via the MN. Keep in mind that most vegetated ecosystems involve mycorrhizae, not just forests. Interestingly, the hyphae that penetrate, surround, and spread over large distances between colonised plants form a large portion of potential grassland carbon sequestration.

Two dominant groups of fungi form MN – arbuscular and ectomycorrhizal fungi. Both types function similarly but have distinctive structures. The hyphae of arbuscular mycorrhizal fungi (AMF) penetrate cortical (outer) root cells and form branching structures within the cytoplasm called *arbuscules* for nutrient exchange and vesicles for storage. Ectomycorrhizal fungi (EMF) create a sheath (known as the mantle) around the roots, forming an interface of hyphae surrounding individual root cells extending from the inner mantle called a 'Hartig net.'





LTR: Fungal Structures Associated With Plant Roots, Arbuscules and Vesicles Within Cortical Cells

These distinctions are important on an ecosystem scale as, generally, individuals who partner with EMF are often fitter than those who partner with AMF. The protective mantle of EMF provides a physical barrier to pathogens. Instead, AMF directly penetrate the root, creating an entry point for pathogens. Yet why do AMF associate with 80% (Figueiredo, Boy and Guggenberger, 2021) of land plant species whereas EMF only associate with 1.7%*? The answer requires a bit of ecology 101.

Biodiversity arises as competition between and within species to survive and reproduce drives natural selection and, over time, evolution. When a forest ecosystem reaches a climax community – the most complex it can be – a few dominant species (often the largest tree species) 'control' the ecosystem. However, ecosystems that partner with AMF experience negative feedback of nutrients between plants and the soil, and expose dominant trees to root damage as pathogens can hack the root firewall. This allows other species to swoop in and claim a portion of the niche (role) the dominant species can no longer fulfil to the same extent. Therefore, a greater species richness leads to a healthier ecosystem overall, explaining why AMF is 'favoured' over EMF. Still, ectomycorrhizae are common in temperate or boreal forests where intraspecific genetic diversity and collaboration may contribute more to ecosystem stability than species diversity.

Article By OCN

* The percentage value for EMF was calculated using data for the number of plant species that are associated with EMF from Science Direct (2022) and the total number of known plant species from Royal Botanic Gardens, Kew (2023).

Recommendations

A-Level Wider Independent Study

- **Mathematics:** What Is Bayes Theorem?
- **Physics:** The Physics of Pentaquarks.
- **Chemistry:** The Chemistry of Element 119.
- **Biology:** Digestion of Lipids.

Discrete Further Reading

- **Mathematics:** *Quanta Magazine September 2024*, Mathematicians Discover New Shapes to Solve Decades-Old Geometry Problem
- **Physics:** *APS Physics Magazine September 2024*, Positron Emission Tomography Could Be Aided by Entanglement
- **Chemistry:** *Nature September 2024*, Carbon bond that uses only one electron seen for first time: 'It will be in the textbooks'
- **Biology:** *Nature September 2024*, How your brain detects patterns in the everyday: without conscious thought

Continuous Further Study Topics

- **Mathematics:** How Does Recorde's Multiplication Method Work?
- **Physics:** What Are Bosons?
- **Chemistry:** The Forms of Elemental Sulfur
- **Biology:** Alphafold AI Modelling of Proteins

Book Appraisal

Moalem, S. and Prince, J. (2008). *Survival of the Sickest : The Surprising Connections Between Disease and Longevity.* New York: Harper Perennial.

Monthly Problem

Solve for x:

$$\frac{9^{2x+1} \times 3^{4-3x}}{27^{2-x}} = 9$$

Solution next month.

> **Last month's solution:**
> $x = 4$

Problem By NSR, Suggestions By NSR & OCN

Credits

Sources & Provenance

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Corrections

No new errors have been identified within previous issues of this newsletter. The editor can be contacted regarding errors and suggestions via the email skinners-science-spectrum@magna-aspirations.org.

Join us next month for some chemistry papercraft, now, where did I put my paper…

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